

Note su impulsi e distribuzioni temperate

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Generalità sugli impulsi

- ❖ *In matematica, la funzione delta di Dirac (δ) è una funzione o distribuzione generalizzata introdotta da Paul Dirac.*
- ❖ *Si usa per modellare la densità di una massa o carica di punto ideale come funzione pari a zero dappertutto a parte lo zero e il cui integrale sulla linea reale è pari a uno.*
- ❖ *Chiaramente non esiste funzione con tali proprietà, quindi i calcoli eseguiti dai fisici teorici sono apparsi assurdi ai matematici fino alla definizione delle distribuzioni di Laurent Schwartz per formalizzare e validare i calcoli.*
- ❖ *Essendo una distribuzione, la funzione delta di Dirac è un operatore lineare che mappa ogni funzione, continua in un intorno dello zero, al suo valore a zero.*
- ❖ *In ingegneria e nelle tecnica di elaborazione dei segnali, la funzione delta, detta anche impulso unitario, si può utilizzare tramite la sua trasformata di Fourier o di Laplace, pensata come limite di una funzione analitica complessa di variabile complessa.*
- ❖ *Questa funzione e le sue trasformate sono parte del calcolo operativo, prezioso strumento di fisica e ingegneria.*
- ❖ *Spesso, delta di Dirac è considerata una specie di limite (limite debole) di una sequenza di funzioni con picco elevato all'origine (in teoria delle distribuzioni, è un vero limite). Le funzioni approssimative della sequenza sono funzioni delta "approssimative».*

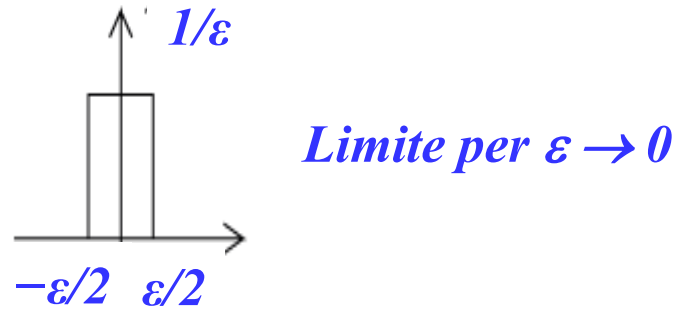
Generalità sugli impulsi

- ❖ *Relativamente alla necessità di considerare le funzioni impulsive (Delta di Dirac) nella tecnica della trasformata di Fourier, è bene riflettere che il più grande svantaggio della classica trasformazione di Fourier è che esiste una classe piuttosto ristretta di funzioni per le quali può essere calcolata classicamente.*
- ❖ *Infatti, ad esempio, è necessario che queste funzioni diminuiscano sufficientemente rapidamente fino a zero (tendendo all'infinito) per garantire l'esistenza dell'integrale di Fourier.*
- ❖ *Così, la trasformata di Fourier di funzioni semplici come i polinomi definiti sull'intero asse reale dei tempi non esiste in senso classico e nemmeno della funzione a gradino.*
- ❖ *Si può obiettare che la trasformata di Laplace può essere applicata in molti di questi casi, e in molti altri ancora, ma così di fatto si perde la possibilità di passare al dominio della frequenza, così prezioso per le implicazioni fisiche (anche se il concetto di frequenza è, di per sé, in effetti fisicamente sfuggente).*
- ❖ *L'estensione della classica trasformazione di Fourier alle distribuzioni ha ampliato considerevolmente la classe di funzioni che potevano essere trasformate e ciò ha rimosso molti ostacoli.*

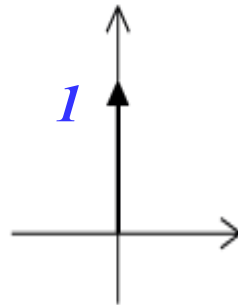
Dirac Delta Function

- ❖ *La funzione delta di Dirac, spesso detta impulso o delta function, è la (successione di) funzioni che concretizza l'idea di impulso unitario in condizioni di tempo continuo.*
- ❖ *Informalmente, questa funzione è infinitamente stretta attorno allo 0 (o altro valore a), infinitamente elevata in tale intervallo infinitesimo, con integrale unitario.*
- ❖ *Il più semplice modo di visualizzare tale comportamento (però non utile per le derivate degli impulsi) è un impulso rettangolare definito tra $(a)-\varepsilon/2$ e $(a)+\varepsilon/2$ con valore $1/\varepsilon$ in tale intervallo.*
- ❖ *Procedendo al limite, ovvero considerando il limite di questo insieme di funzioni per ε che tende a 0, l'ampiezza dell'intervallo di definizione tende a zero ed il valore degli impulsi in tale intervallo tende a infinito. con area totale che rimane costante a uno.*
- ❖ *La funzione impulsiva è spesso scritta come $\delta(t)$ e per essa $\int_{-\infty}^{\infty} \delta(t) dt = 1$.*
- ❖ *La trattazione che segue è assolutamente informale, dati gli obiettivi del corso.*

Dirac Delta Function – una semplicissima approssimazione



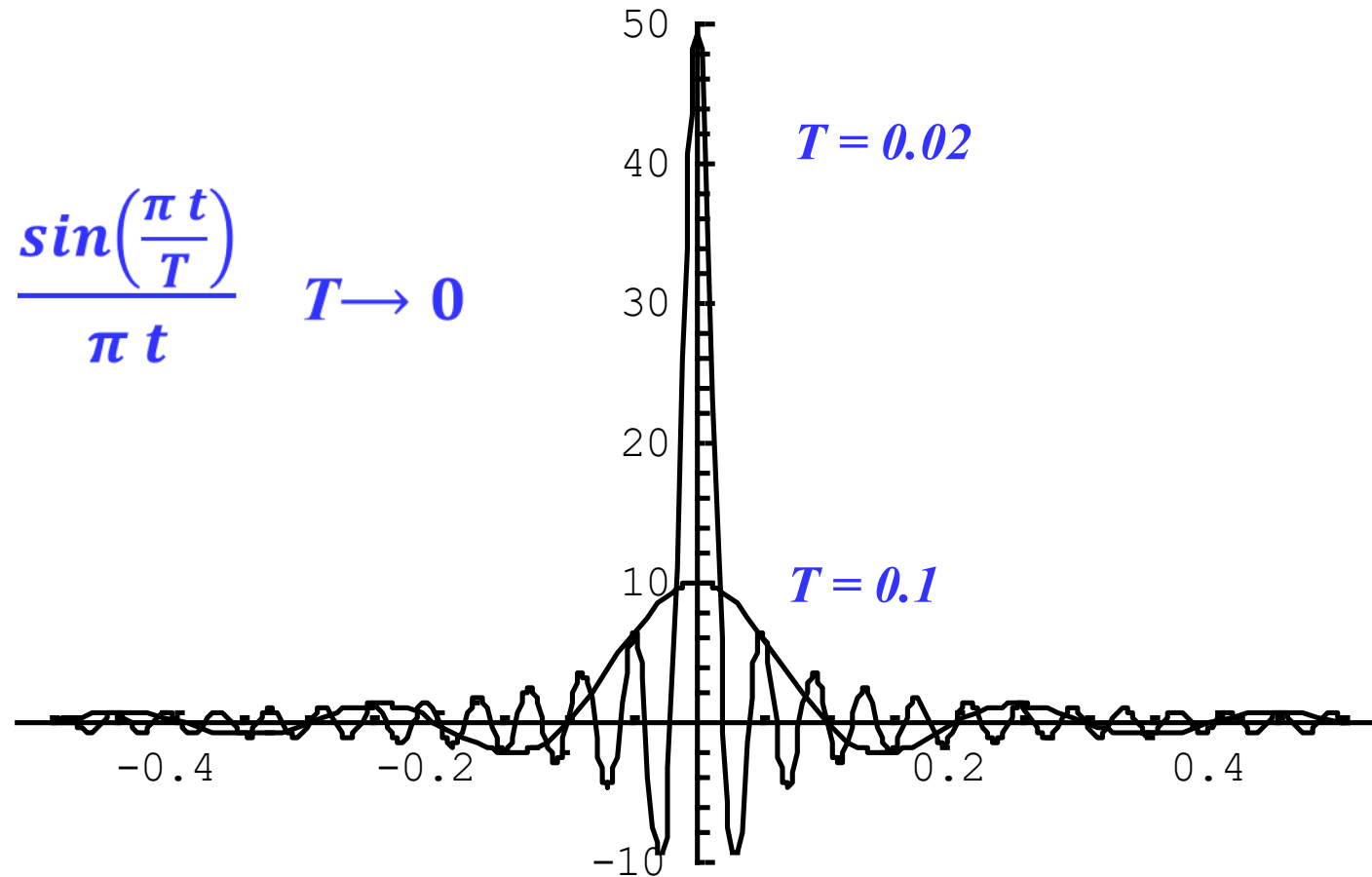
Ecco un modo semplice per visualizzare la funzione Delta di Dirac



Poiché è difficile disegnare qualcosa che è infinitamente elevato, si rappresenta la funzione Delta di Dirac con una freccia centrata nel tempo ove è applicata. Il valore (integrale) degli impulsi (1 o altri valori) è indicato vicino al vertice della freccia.

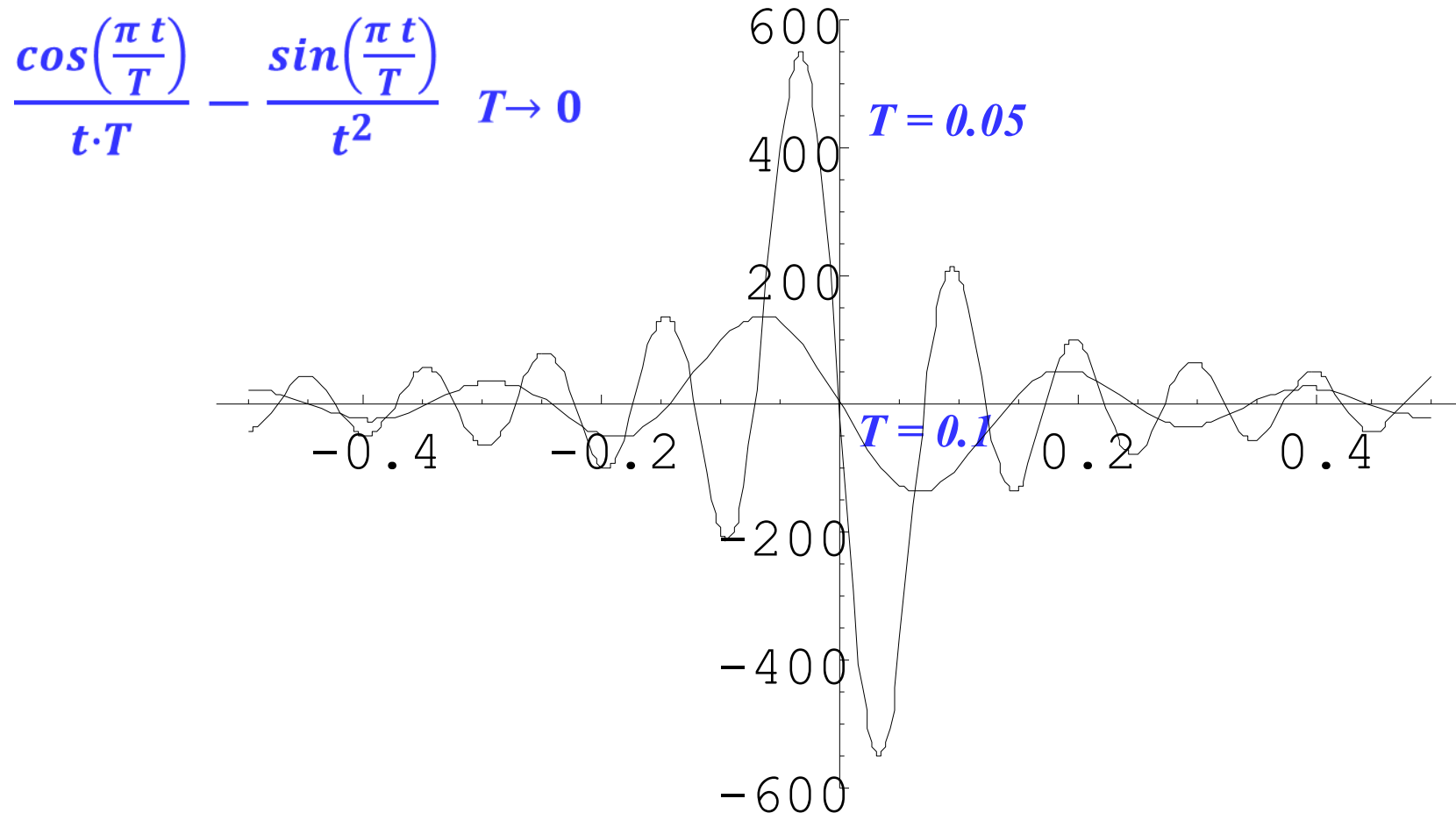
Dirac Delta Function – una migliore approssimazione

Più efficacemente si può considerare la successione di funzioni sinc ad area unitaria



Dirac Delta Function – una migliore approssimazione

Il vantaggio è che si possono visualizzare le derivate della funzione approssimata



Dirac Delta Function – proprietà

Proprietà dell'impulso unitario

$$\bullet \delta(a \cdot t) = \frac{\delta(t)}{|a|} \quad a \neq 0$$

• *dalla proprietà precedente $\delta(t) = \delta(-t)$ (pari)*

• $\delta(t) = \frac{du(t)}{dt}$, *ove $u(t)$ è il gradino unitario. infatti $u(t) = \int_{-\infty}^t \delta(\tau) d\tau$*

$$\bullet f(t) \cdot \delta(t) = f(0) \cdot \delta(t)$$

L'ultima di queste è specialmente importante perché origina la proprietà di shifting della funzione Delta di Dirac, che seleziona il valore di una funzione ad un tempo specifico ed è soprattutto importante nello studio dell'operatore di convoluzione nella analisi temporale dei sistemi lineari tempo invarianti.

La proprietà precedente è illustrata (informalmente) di seguito.

$$\int_{-\infty}^{\infty} f(t) \cdot \delta(t) dt = \int_{-\infty}^{\infty} f(0) \cdot \delta(t) dt = f(0) \cdot \int_{-\infty}^{\infty} \delta(t) dt = f(0)$$

Dirac Delta Function – proprietà

❖ *Poiché*

$$\int_{-\infty}^{\infty} t \cdot \delta(t) dt = f(0) = 0$$

allora $t \cdot \delta(t) = 0$

❖ *Poiché, del tutto informalmente, $\delta(t-a)$ è definita in un intorno infinitamente piccolo nell'intorno di a , ove l'argomento della Delta di Dirac si annulla, sembra del tutto accettabile che $\delta(g(t))$ sia definito solo negli intorni degli zeri di $g(t)$ e si abbia*

$$\delta(g(t)) = \sum_k \frac{\delta(t - x_k)}{|g'(x_k)|}$$

x_k sono le radici (singole) di $g(x)$, supposta continua e derivabile in corrispondenza di tali radici; i denominatori nella relazione precedente sono informalmente comprensibili poiché negli intorni delle radici si ha $f(t) \cong f'(x_k) \cdot (t - x_k)$ e se si considerano le proprietà precedenti.

❖ *Dalla relazione precedente, sempre informalmente, si ha subito*

$$\int_{-\infty}^{\infty} f(t) \cdot \delta(g(t)) dt = \sum_k \frac{f(x_k)}{|g'(x_k)|}$$

Derivate della funzione impulsiva

Derivata prima della funzione impulsiva $\delta'(t)$ per cui si ha

$$\int_{-\infty}^{\infty} \delta'(t) \cdot t dt = \delta(t) \cdot t \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \delta(t) dt = - \int_{-\infty}^{\infty} \delta(t) dt$$

il termine valutato è nullo per la definizione di $\delta(t)$, quindi informalmente.

$$\delta'(t) = - \frac{\delta(t)}{t}$$

Proseguendo con le derivate si ottiene

$$\int_{-\infty}^{\infty} \delta^n(t) \cdot t^n dt = (-1)^n n! \int_{-\infty}^{\infty} \delta(t) dt = \quad \Rightarrow \quad \delta^n(t) = (-1)^n n! \frac{\delta(t)}{t^n}$$

$$\delta^n(-t) = (-1)^n \delta^n(t)$$

Derivate della funzione impulsiva - proprietà

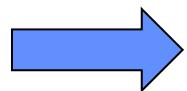
Proprietà della derivata prima della funzione impulsiva $\delta'(t)$

$$\int_{-\infty}^{\infty} \delta'(t-a) \cdot f(t) dt = \delta(t-a) \cdot f(t) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \delta(t-a) \cdot f'(t) dt = -f'(a)$$

*Ove $f(t)$ è continua in $t=a$ con la derivata prima
il termine valutato è nullo per la definizione di $\delta(t)$.*

Proseguendo con le derivate si ottiene

$$\int_{-\infty}^{\infty} \delta^n(t) \cdot f(t) dt = - \int_{-\infty}^{\infty} \delta^{n-1}(t) \cdot f'(t) dt$$



$$\int_{-\infty}^{\infty} \delta^n(t) \cdot f(t) dt = (-1)^n f^n(0)$$

Trasformata di Fourier della funzione impulsiva

Per la trasformata di Fourier della funzione impulsiva $\delta(t)$ si ha

$$\mathfrak{F}[\delta(t-a)] = \int_{-\infty}^{\infty} \delta(t-a) \cdot e^{-j2\pi f t} dt = e^{-j2\pi f a}$$

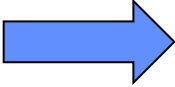

$$\mathfrak{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j2\pi f t} dt = 1$$

La trasformata di Fourier di $\delta(t)$ è la costante 1 in f (o la sua traslazione temporale), per la dualità la trasformata di Fourier della costante 1 nel tempo è $\delta(f)$, $\mathfrak{F}[\delta(t)] = 1$, $\mathfrak{F}[1] = \delta(f)$.

Proseguendo con le derivate si ottengono trasformate e antitrasformate per t^n e f^n

$$\mathfrak{F}[\delta^n(t)] = \int_{-\infty}^{\infty} \delta^n(t) \cdot e^{-j2\pi f t} dt = (-1)^n \frac{d^n e^{-j2\pi f t}}{dt^n} \Big|_{t=0} = (j2f\pi)^n$$

dualità $\delta^n(-f) = (-1)^n \delta^n(f) \Leftrightarrow (j2t\pi)^n \qquad \delta^n(t) \Leftrightarrow (j2f\pi)^n$


$$\left(\frac{j}{2\pi}\right)^n \delta^n(f) \Leftrightarrow (t)^n$$

Impulsi e derivate moltiplicati per funzioni di f

- ❖ *Con impulsi si ottiene con facili considerazioni $\delta(f) \cdot F(f) = \delta(f) \cdot F(0)$.*
- ❖ *Per la derivata $\delta'(f) \cdot F(f) = [-\delta(f)/f] \cdot F(f) = [-\delta(f)/f] \cdot [F(0) + F'(0) \cdot f] = \delta'(f) \cdot F(0) - \delta(f) \cdot F'(0)$.*
- ❖ *E così per le derivate successive, ammesso che $F(f)$ sia sviluppabile con Taylor.*

Trasformata di Fourier del gradino unitario

Si consideri la trasformata di $u(t)$ come limite

$$\mathfrak{F}[u(t)] = \lim_{T \rightarrow \infty} \int_0^T e^{-j 2 \pi f t} dt = \lim_{T \rightarrow \infty} \frac{1 - e^{-j 2 \pi f T}}{j 2 \pi f} =$$
$$\frac{1}{j 2 \pi f} + \lim_{T \rightarrow \infty} \frac{-e^{-j 2 \pi f T}}{j 2 \pi f} = \frac{1}{j 2 \pi f} + \lim_{T \rightarrow \infty} \frac{-\cos(2 \pi f T) + j \sin(2 \pi f T)}{j 2 \pi f} =$$

$$\lim_{T \rightarrow \infty} \frac{-\cos(2 \pi f T)}{j 2 \pi f} = 0$$

$$\lim_{T \rightarrow \infty} \frac{\sin(2 \pi f T)}{2 \pi f} = \frac{\delta(f)}{2}$$

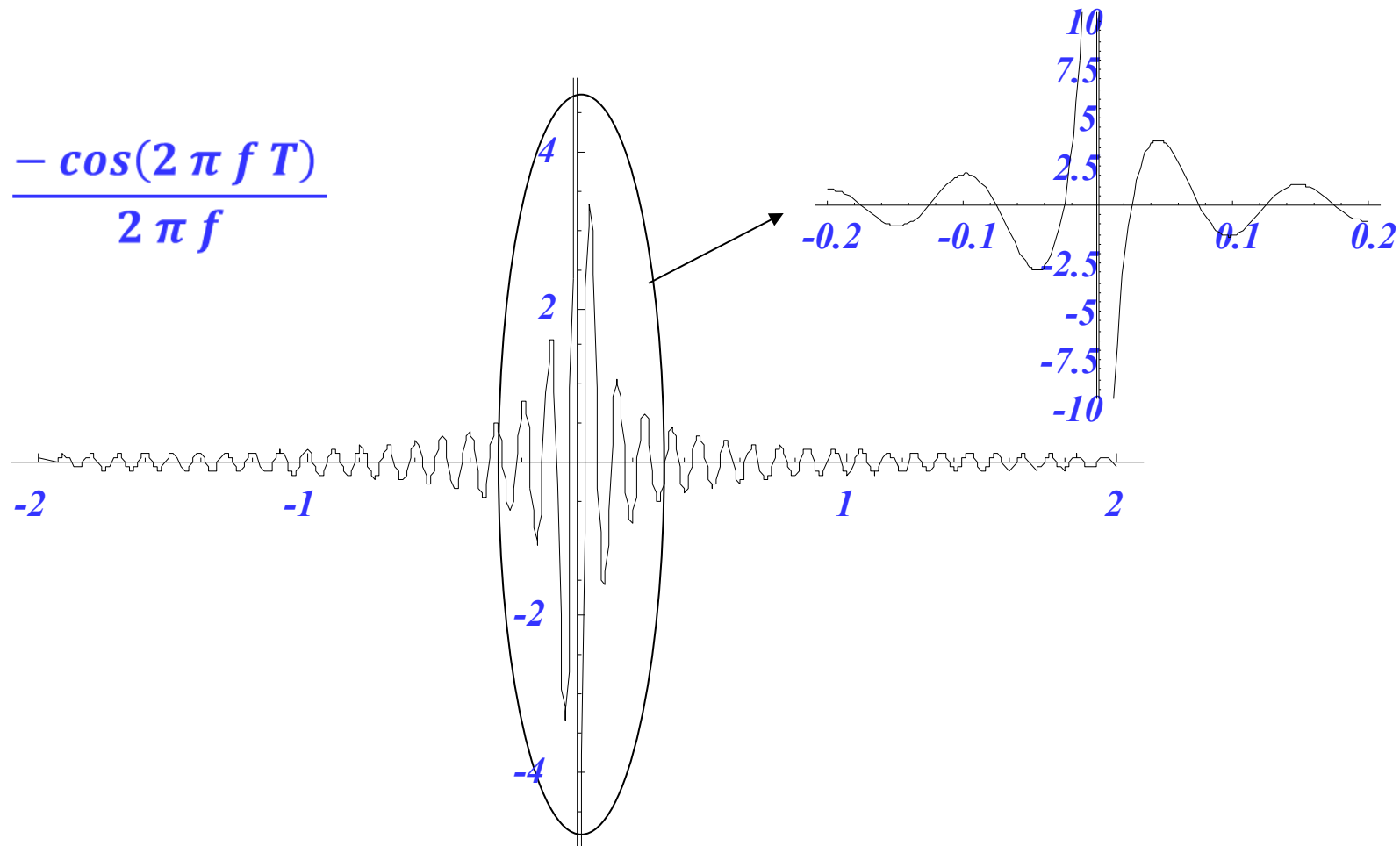
$$\mathfrak{F}[u(t)] = \frac{1}{j 2 \pi f} + \frac{\delta(f)}{2}$$

È una funzione dispari, con integrale nullo su qualsiasi intervallo di tempo e anche per intervalli di tempo simmetrici attorno allo 0 se $T \rightarrow \infty$, integralmente è una funzione nulla

Approssimazione della funzione impulsiva se $T \rightarrow \infty$

L'integrale della funzione tra $-\infty$ e ∞ vale 0.5

Trasformata di Fourier del gradino unitario



Trasformata di Fourier della funzione segno

Si consideri la trasformata di $\text{sign}(t)$ come limite

$$\begin{aligned}\mathfrak{F}[\text{sign}(t)] &= \lim_{T \rightarrow \infty} \int_{-T}^T \text{sign}(t) \cdot e^{-j 2 \pi f t} dt = \lim_{T \rightarrow \infty} \frac{-2j \sin(\pi f T)^2}{\pi f} = \\ &= \lim_{T \rightarrow \infty} \frac{1 - \cos(\pi f T)}{j \pi f} = \\ \lim_{T \rightarrow \infty} \frac{-\cos(\pi f T)}{j \pi f} &= 0 \quad \text{Come già osservato, integralmente è una funzione nulla}\end{aligned}$$

$$\mathfrak{F}[\text{sign}(t)] = \frac{1}{j \pi f} \quad \text{Per la dualità} \quad \text{sign}(-f) = -\text{sign}(f) \Leftrightarrow \frac{1}{j \pi t}$$

$$\mathfrak{F}\left[\frac{1}{t}\right] = -j \pi \text{sign}(f)$$

Fourier transform of tempered distributions

Functions and distributions

- ❖ *The main difference between functions and distributions lies in the fact that an ordinary function has a precise value at every point, whereas a distribution (or generalized function) is defined only through the way it "acts" on other functions.*
- ❖ *Here are the key differences:*
 - ❖ *1. Definition Ordinary function: It is a rule that associates to each number x a number y . For example, $f(x) = x^2$. You can calculate its pointwise value: $f(2) = 4$. Distribution: It is an "operator" (technically a continuous linear functional). It does not have a value at a point, but produces a number when it is "tested" with a very regular function, called a test function.*
 - ❖ *2. Evaluation Function: You evaluate it at the point: $f(x)$. Distribution: You evaluate it through an integral (real or symbolic). If T it is a distribution and ϕ a test function, the result is indicated as $\langle T, \phi \rangle$*
 - ❖ *3. Differentiability Ordinary functions: Not all are differentiable (think of a function with a jump or a sharp corner). Distributions: They are always infinitely differentiable. The result of a derivative that would not exist in the classical sense (like the derivative of a step function) produces a new distribution (like the Dirac Delta).*

Functions and distributions

- ❖ *4. Classic Example: The Dirac Delta (δ). The Dirac Delta is the perfect example of a distribution that is not an ordinary function: If it were a function, it would have to be 0 everywhere except at 0, where it would have to be "infinite." But such a function would have a zero integral. As a distribution, its definition is very clear: when it meets a test function ϕ , it extracts its value at the origin: $\langle \delta, \phi \rangle = \phi(0)$.*
- ❖ *In short: Distributions are an extension of the concept of function that allows handling "irregular" mathematical objects (impulses, sharp jumps) with the same rigor and the same ease of calculation as normal functions.*

Functions and distributions

- ❖ *It is possible that ordinary functions and distributions coincide, or more precisely, that an ordinary function generates a distribution (regular distribution).*
- ❖ *Here are the key points:*
- ❖ *Regular Distributions: A function $f(x)$ that is locally summable (i.e., integrable over every compact interval) can be seen as a distribution T_f that acts on a test function ϕ in the following way:*

$$\langle T, \phi \rangle = \int_{-\infty}^{\infty} f(x) \cdot \phi(x) dx$$

In this sense, the function $f(x)$ and the distribution T_f are identified.

- ❖ *Identification: When talking about distributions that "coincide" with functions, it means that the distribution is generated by a continuous or locally integrable function.*
- ❖ *Difference: Unlike singular distributions (like the Dirac delta), which cannot be represented as standard functions, regular distributions behave like the functions they come from.*
- ❖ *So, every continuous function is a distribution, but not all distributions are functions (for example, the Dirac delta is not an ordinary function)*

Notes on temperate distributions

- ❖ *For its applications to Fourier transforms that cannot otherwise be calculated, we briefly recall the basic characteristics of tempered distributions. A continuous and locally integrable function $f(t)$ (integrable over every compact subset (closed (includes its accumulation points) and bounded), but not integrable in the proper sense, for example $1/t \forall t$ is locally integrable, but not on intervals that include 0) defines a tempered distribution if and only if it has polynomial growth at infinity, that is, if there exists a k such that $|f(t)| \leq C(1+|t|^k)$; examples include polynomials and the Dirac delta and its derivatives.*
- ❖ *Therefore, while classical distributions can grow exponentially, tempered distributions do not grow faster than a polynomial at infinity.*
- ❖ *Tempered distributions allow the extension of the Fourier transform to non-integrable objects, such as constant functions, polynomials, or of the type t^{-n} , as seen previously.*
- ❖ *For example, the Fourier transform of $f(t) = 1/t^2 \forall t$, which obviously satisfies the given conditions, is given by $\mathcal{F}[f(t)] = -2\pi^2 f \text{ sign}(f)$.*
- ❖ *In fact, the given function is the derivative, in the sense of the principal value, of $-1/t$, that is $1/t^2 = -d/dt (1/t)$, and since the transform of $1/t$ is $-\pi j \text{ sign}(f)$, then $\mathcal{F}[f(t)] = -j 2\pi f - \pi j \text{ sign}(f) = -2\pi^2 f \text{ sign}(f)$.*

Fourier transform of t^α with rational α

❖ *Nella parte dedicata agli impulsi di Dirac si è trovato che*

$$\mathcal{F}[t^n] = \mathcal{F}[t^n \cdot 1(t)] \text{ } n \text{ intero positivo è } 1/(-j2\pi)^n \delta^{(n)}(f)$$

❖ *ove $\delta^{(n)}(f)$ è la derivata n -esima di $\delta(f)$.*

❖ *Generalizzando e considerando le derivate frazionali*

$$\mathcal{F}[t^x] = \mathcal{F}[t^x \cdot 1(t)] \text{ } x \text{ razionale } > 0 \text{ è } 1/(-j2\pi)^x \delta^{(x)}(f)$$

❖ *ove $\delta^{(x)}(f)$ con x razionale > 0 è $f^{-x-1} / \Gamma(-x)$*

❖ *Essendo $\Gamma(\alpha)$ la funzione Gamma-*

❖ *Per α intero $\Gamma(\alpha) = (\alpha-1)!$*

$$\Gamma(1/2) = \sqrt{\pi}$$

$$\Gamma(-1/2) = -2 \Gamma(1/2) = -2\sqrt{\pi}$$

$$\Gamma(1/2 - n) = (n)! (-4)^n \sqrt{\pi} / (2n!) \text{ } n \text{ intero non negativo}$$

$$\Gamma(1/2 + n) = (2n)! \sqrt{\pi} / (4^n n!) \text{ } n \text{ intero non negativo}$$

❖ *Analogamente, $\mathcal{F}[t^{-x}]$ x razionale > 0 è $\Gamma(1-x) (j2\pi f)^{x-1}$*

Insights on tempered distributions

Notes on temperate distributions

- ❖ *Tempered distributions ($\mathcal{S}'(\mathcal{R})$) are a specific class of generalized functions that act as linear and continuous functionals on the Schwartz space $\mathcal{S}(\mathcal{R})$ (rapidly decreasing functions). They can be thought of as functions with "slow growth" at infinity. Thus, since the distributions are slowly increasing to infinity while the functions of Swartz are rapidly decreasing, the functional $\langle T, \phi \rangle$ certainly exists. Here are examples:*
- ❖ *1. Classical Periodic Functions*
- ❖ *Constant functions: $f(x) = c$ are the simplest periodic distributions. Any function that is periodic ($\sin(x)$, $\cos(x)$) is a bounded periodic tempered distribution. Locally integrable functions that are periodic (for example, a square wave or a sawtooth wave).*
- ❖ *2. Polynomials (t^n)*
- ❖ *Any polynomial function $P(t) = a_n t^n + \dots + a_0$ is a tempered distribution.*
 - ❖ *Example: $P(t) = t^2 + 3t + 1$*
- ❖ *3. Inverse/ Singular Power Functions (t^{-n})*
- ❖ *Inverse powers are not well-defined at the origin as classical functions, but they can be as distributions (principal distributions or derivatives).*
 - ❖ *Cauchy's Principal Value: $vp(1/t)$ is a tempered distribution.*

Notes on temperate distributions

- ❖ *4. Impulses and Singular Distributions*
- ❖ *Impulses and their derivatives are the most common tempered distributions.*
 - ❖ *Dirac Delta: $\delta(t-a)$ (impulse in $t = a$).*
 - ❖ *Derivative of the impulse: $\delta'(t)$.*
 - ❖ *Heaviside step function: $H(t) = 1$ per $t > 0$
 $= 0$ per $t < 0$*
- ❖ *The Fourier transform is a topological isomorphism of both the Schwartz space ($\mathcal{S}(\mathcal{R}^n)$) and the space of temperate distributions ($\mathcal{S}'(\mathcal{R}^n)$). In both cases, the transform transforms space into itself (or into its dual, in the case of ($\mathcal{S}'(\mathcal{R}^n)$), is linear, continuous, invertible, and with continuous inverse.*
- ❖ *Tempered distributions are always Fourier transformable because they are defined as linear and continuous functionals on the Schwartz space ($\mathcal{S}(\mathcal{R}^n)$), a space of rapidly decreasing functions. The Fourier transform is an automorphism on this space, it transforms Schwartz functions into themselves, allowing the operation to be extended to distributions by duality.*

Cenni alle distribuzioni temperate

- ❖ *The Fourier transform behaves exceptionally in the Schwartz space $\mathcal{S}$, i.e. the space of infinitely derivable functions with rapid decrease. Anyway, the crucial requirement of Schwartz space is not only decay, but unlimited regularity. A cusp, a corner, a derivative discontinuity — even just one — destroys membership in $\mathcal{S}$.*
- ❖ *Here are the key points:*
- ❖ *On the $\mathcal{S}$: It is an automorphism (an isomorphism from $\mathcal{S}$ in itself). Because of the derivation and multiplication properties for the polynomial, if a function is in $\mathcal{S}$, its transform is also in $\mathcal{S}$. It is a linear, continuous and two-way operator.*
- ❖ *On the dual $\mathcal{S}'$: Since the transform is an isomorphism on $\mathcal{S}$, it can be extended by duality to the space of the temperate distributions ($\mathcal{S}'$). Also in this case it is an isomorphism: this allows to transform objects that would not be classically treatable, such as polynomials or delta of Dirac.*
- ❖ *In synthesis, the structure of $\mathcal{S}$ is "made on purpose" to make the Fourier transform a perfect and symmetric operation.*
- ❖ *Similarly, the Fourier transform is a topological isomorphism (or automorphism) of the space of tempered distributions $\mathcal{S}'(\mathcal{R}^n)$ onto itself*

Notes on temperate distributions

- ❖ *The proof of why the Fourier transform always exists for a temperate distribution ($T \in \mathcal{S}'(\mathcal{R}^n)$) is based on duality.*
- ❖ *Here are the logical steps:*
- ❖ *For a "normal" (integrable) function (f), the relation applies:*
$$\langle f^\wedge, \phi \rangle = \langle f, \phi^\wedge \rangle \quad \text{with } x^\wedge \text{ denoting the Fourier transform of the function } x$$
- ❖ *where ϕ is a Schwartz function. Since we want the transform to be consistent with the classical case, we define the Fourier transform of a tempered distribution (T) as*
$$\langle T^\wedge, \phi \rangle := \langle T, \phi^\wedge \rangle \text{ it is sufficient to swap the order of integration}$$
- ❖ *The crucial point is that the classical Fourier transform $\mathcal{F}$ is an isomorphism (a linear and continuous operator) of the Schwartz space $\mathcal{S}(\mathcal{R}^n)$ into itself:*
$$\phi \in \mathcal{S} \rightarrow \phi^\wedge \in \mathcal{S}$$
- ❖ *If we take a function that decays rapidly (like ϕ), its transform will also decay rapidly.*
- ❖ *By definition, a tempered distribution (T) is a linear and continuous functional on $\mathcal{S}$. Since the operation $\phi \rightarrow \phi^\wedge$ is continuous within $\mathcal{S}$, the composition $\langle T, \phi^\wedge \rangle$ remains a linear and continuous functional.*
- ❖ *Conclusion: Since $\phi^\wedge$ still belongs to $\mathcal{S}$ for every ϕ , the expression $\langle T, \phi^\wedge \rangle$ always has a well-defined value. As a result, $T^\wedge$ is itself a tempered distribution.*

Notes on temperate distributions

- ❖ *The Fourier transforms and inverse transforms are considered an isomorphism (specifically, between spaces $\mathcal{L}^2$) because they establish a one-to-one correspondence that preserves the entire mathematical structure of signals.*
- ❖ *Here are the main reasons in simple words:*
 - ❖ *Linearity: If you add two signals or multiply them by a constant, their transform reflects exactly the same operation. Then, the algebraic structure remains.*
 - ❖ *Bijectivity (Invertibility): Each signal in the time domain corresponds to one and only one representation in the frequency domain, and vice versa. The inverse transform is the perfect "return trip": you do not lose information in the process.*
 - ❖ *Preservation of the Inner Product (Parseval/Plancherel Theorem): This is the fundamental part for defining a unitary isomorphism. The signal's energy (its norm) is identical both in time and frequency. The operator "preserves distances".*
 - ❖ *Transformation of Operations: Complicated things like convolution over time become simple multiplications in frequencies. This "transfers" the properties of computation from one environment to another without breaking them.*
- ❖ *In short, for mathematics, the time domain and the frequency domain are the same space seen from two different perspectives; the transform is simply the "switch of glasses" that allows you to go from one to the other.*

Notes on temperate distributions

- ❖ *The derivative in the sense of the principal value is a mathematical tool used to define the derivative at points where, according to the classical definition, it would not exist, yet there is a sort of symmetry in the behavior of the function.*
- ❖ *For example, the derivative of $\text{sign}(t)$ at 0 is thus $2 \delta(t)$; the derivative of $1/t^k$, k an integer, also at 0, is calculated by excluding a small interval $[-\varepsilon, \varepsilon]$ around 0 and taking the limit as $\varepsilon \rightarrow 0$ of the resulting finite part (obtaining $-k/t^{(k+1)}$).*
- ❖ *By simple induction, based on $\mathcal{F}[1/t]$, $\mathcal{F}[1/t^{2n+1}]$ (n integer) $= -j \pi (2\pi f)^{2n} \text{sign}(f) / (2n-1)!$ e $\mathcal{F}[1/t^{2n}]$ (n integer) $= -\pi |2\pi f|^{2n-1} / (2n-1)!$*
- ❖ *Regarding the properties of the Fourier transform, the transform of the derivative of a function is $j2\pi f$ times the transform of the function and, unlike the classical case, it is not necessary for the function to be actually differentiable and to decay at infinity; for tempered distributions, the formula is always valid.*
- ❖ *Similarly, the transform of $t \cdot f(t)$ is $(1/(-j2\pi)) d/df \mathcal{F}[f(t)]$, where the derivative is to be understood in the sense of distributions, even if the $\mathcal{F}[f(t)]$ it is not derivable in the classical sense.*

Cauchy principal value

- ❖ *The Cauchy principal value (often denoted as P.V. $\int$ or v.p. $\int$) is a method for assigning a finite value to improper integrals that would otherwise diverge, by balancing the singularities. It is calculated as the limit of the sum of two integrals that approach a singularity symmetrically, that is:*

$$P.V. = \int_a^b f(x) \, dx = \lim_{\varepsilon \rightarrow 0^+} \left(\int_a^{x_0 - \varepsilon} f(x) \, dx + \int_{x_0 + \varepsilon}^b f(x) \, dx \right)$$

- ❖ *Here are the key points:*
- ❖ *Management of singularities: It is used when a function $f(x)$ becomes infinite at a point x_0 inside the interval $[a,b]$, making the integral improper according to the standard definition.*
- ❖ *Symmetric approach: Unlike the classical improper integral, the Cauchy principal value requires approaching the singularity with the same radius ε from the right and from the left, often allowing the 'cancellation' of infinite contributions of opposite sign.*
- ❖ *Integrals over infinite domains: It can also be applied to integrals over the entire real axis $(-\infty, +\infty)$, defined as $\lim_{R \rightarrow \infty} \left(\int_{-R}^R f(x) \, dx \right)$.*

Cauchy principal value - an example

- ❖ *The Fourier transform of the Cauchy principal value of t^{-1} , defined in the sense of distributions, is: $\mathcal{F}\{\text{v.p.}\{1/t\}\} = -j \pi \text{sgn}(2 \pi f)$.*
- ❖ *1. Definition of the principal value distribution*
- ❖ *The function $f(t) = 1/t$ is not integrable in the strict sense on $\mathcal{R}$ due to the singularity in $t = 0$. To calculate its transform, the Cauchy Principal Value is used, defined as the limit for $\varepsilon \rightarrow 0^+$ of the integral that excludes the interval $[-\varepsilon, \varepsilon]$:*
$$\langle \text{v.p. } 1/t, \phi(t) \rangle = \lim_{\varepsilon \rightarrow 0^+} \left(\int_{|t| > \varepsilon} \phi(t) \cdot 1/t \, dt = \int_0^\infty [\phi(t) - \phi(-t)] \cdot 1/t \, dt \right)$$
- ❖ *This expression is equivalent to considering the local behavior $[\phi(t) - \phi(0)] \cdot 1/t$ to ensure convergence near zero.*
- ❖ *2. Calculation using the sign function $\text{sgn}(t)$. To find the Fourier transform, the duality with the sign function is used. We know that the Fourier transform of $\text{sgn}(t)$ is: $\mathcal{F}\{\text{sgn}(t)\} = 1/(j \pi f)$.*
- ❖ *By applying the duality theorem $\mathcal{F}\{\mathcal{F}[f(t)]\} = 2\pi f(-2\pi f)$, the result for $1/t$ is $-j\pi \text{sgn}(f)$.*

Hadamard finite part

- ❖ *The Hadamard finite part (or partie finie, Pf) is a regularization method that allows assigning a finite value to divergent integrals, specifically "eliminating" the divergent (infinite) terms from the integral.*
- ❖ *Here are the key points:*
- ❖ *Functioning: When an improper integral diverges due to a singularity (the function goes to infinity), the Hadamard finite part extracts the "regular" (finite) part and discards the infinite terms that cause the divergence.*
- ❖ *Application: It is particularly used for integrals containing strong singularities (e.g. $1/x^2$, $1/x^3$), where the classical definition of an improper integral or the Cauchy principal value is not sufficient.*
- ❖ *Significance: Instead of obtaining an infinite value, Hadamard regularization allows the function to be treated as a distribution.*
- ❖ *In summary, it subtracts the divergent part, allowing a meaningful value to be calculated from an integral that would otherwise be indeterminate.*

Hadamard finite part – an example

- ❖ *The Fourier transform of the Hadamard finite part (Pf) of the distribution $1/t^2$ (or t^{-2}), often defined as the derivative in the sense of distributions of $-vp(1/t)$, is related to the absolute value of the angular frequency.*
- ❖ *Using the definition of Fourier transform $\mathcal{F}\{f\}(f) = \int_{-\infty}^{\infty} f(x) \cdot e^{-j2\pi ft} dt$, the result is:*
 $\mathcal{F}\{Pf(1/t^2)\}$
- ❖ *Analysis of the terms*
- ❖ *t^{-2} with Hadamard finite part: Since $1/t^2$ is neither integrable in the Lebesgue sense nor in the principal value (vp) sense in $= 0$, the Hadamard finite part is used to define the distribution. It acts on test functions $\phi(t)$ removing the singularity:*
$$\langle Pf(1/t^2), \phi(t) \rangle = \int_{-\infty}^{\infty} [\phi(t) - \phi(0) - t\phi'(0)] \cdot 1/t^2 dt$$
- ❖ *It is often associated with the definition $-d/dt vp(1/t)$.*
- ❖ *Frequency result: The transform is a continuous and scalar function proportional to $-|2\pi f|$, which has a "cusplike" behavior in $f = 0$.*
- ❖ *The derivative of $vp(1/t)$ ($vp(1/t)$)' is actually $-pv1/t^2$.*

Notes on temperate distributions

- ❖ *The profound reason why the Fourier transform of a rapidly decreasing function (and its derivatives) is still a rapidly decreasing function lies in the duality between the regularity (smoothness) in the time domain and the rate of decay in the frequency domain.*
- ❖ *This concept is formalized in Schwartz space, where the Fourier transform acts as an automorphism (a transformation that maintains the properties of space).*
- ❖ *Here are the basic points:*
- ❖ *Regularity-Decay duality: If a function $f(t)$ is very regular (smooth) and decreases rapidly to infinity, its Fourier transform $f^{\wedge}(f)$ will also be very regular and will decrease rapidly.*
- ❖ *Derivatives and Powers: The Fourier transform transforms the time-domain derivation into a multiplication by $j2\pi f$ in the frequency domain. . Since rapidly decreasing functions are infinitely derivable, decay to infinity of $f(t)$ guarantees that $f^{\wedge}(f)$ decays faster than any power of $1/|f|$.*

Notes on temperate distributions

- ❖ *To demonstrate that the Fourier transform $\mathcal{F}$ maps the Schwartz space $\mathcal{S}$ into itself, we must show that if $x(t) \in \mathcal{S}$ then its transform $x^\wedge(f)$ also satisfies the rapid decay condition:*

$$\sup_f [f^k x^\wedge^{(m)}(f)] < \infty \quad \text{for every non-negative integers } (k) \text{ and } (m).$$

- ❖ *The demonstration is based on the combined use of the two fundamental properties:*
 - ❖ *Derivation property: $\mathcal{F}\{x^{(k)}(t)\} = (j2\pi f)^k x^\wedge(f)$*
 - ❖ *Time multiplication property: $\mathcal{F}\{(-j2\pi t)^m x(t)\} = x^\wedge^{(m)}(f)$*
- ❖ *Let's apply these properties at the same time to isolate the term $f^k x^\wedge^{(m)}(f)$:*
- ❖ *We express the derivative (m)-th:*
- ❖ *We know that $x^\wedge^{(m)}(f)$ is the Fourier transform of the function $g(t) = (-j2\pi t)^m x(t)$. Note that if $x(t) \in \mathcal{S}$, then also $g(t) \in \mathcal{S}$ (because multiplying by a polynomial does not ruin rapid decrease).*
- ❖ *We multiply by f^k :*
- ❖ *for the derivation property, multiplying the transform of $g(t)$ by $(j2\pi f)^k$ is equivalent to transforming the k-th derivative of $g(t)$:*

$$(j2\pi f)^k x^\wedge^{(m)}(f) = \mathcal{F}\left\{ \frac{d^k}{dt^k} [(-j2\pi t)^m x(t)] \right\}$$

Notes on temperate distributions

Conclusion:

The expression inside the curly bracket, let's call it $h(t = d^k/dt^k [(-j2\pi t)^m x(t)])$, is still a function of the Schwartz space (being a linear combination of derivatives and products of rapidly decreasing functions). Since $h(t)$ is absolutely integrable (h in $\mathcal{L}^1$), its Fourier transform is a bounded function.

Thus, the product $f^k x^{(m)}(f)$ is bounded for every (k) and (m) . This proves that $x^{(f)}$ decays faster than any power of $1/f$ and has derivatives that do the same.